Part 1: A Guide to All Necessary Installations – Python & Visual Studio Code (Recommended)

1) Introduction

To start coding with Python, simply downloading the Python package is not enough. For Python to work efficiently, you need the programming language itself as well as an Integrated Development Environment (IDE). The most modern, versatile, and industry-standard way to start programming in Python is by combining the standalone Python installation with **Visual Studio Code (VS Code)**.

This setup offers:

- The pure Python programming language.
- A highly customizable, lightweight, and professional IDE (VS Code).
- The ability to install only the packages you actually need, keeping your system fast and clean.

2) Installation Guide

Step A: Installing Python

1. Download the correct installer for your operating system (Windows, macOS, Linux) from the official website: python.org/downloads.
2. Start the graphical installer by double-clicking the downloaded file.

3. **Crucial Step for Windows:** At the very bottom of the first installation window, make sure to check the box that says **"Add Python to PATH"** before clicking "Install Now". This ensures your computer recognizes Python commands.

Step B: Installing Visual Studio Code

1. Download the VS Code installer for your system from code.visualstudio.com.
2. Run the installer and follow the standard installation prompts. Leave the default settings as they are.

3) Setting Up VS Code for Python

VS Code is like a highly competent word processor specifically designed for code. Out of the box, it is essentially a blank canvas. To make it a powerful Python IDE, you need to install **Extensions** (Add-ons).

Open VS Code, click on the **Extensions icon** on the left sidebar (it looks like four squares), and search for the following highly recommended add-ons:

- **Python (by Microsoft):** This is mandatory. It enables VS Code to understand Python code, run scripts, and provides syntax highlighting.
- **Pylance (by Microsoft):** Usually installed alongside the Python extension. It provides excellent code completion and error checking.
- **Jupyter (by Microsoft):** Allows you to create and run interactive Jupyter Notebooks (.ipynb files) directly within VS Code.
- **Prettier - Code formatter:** Helps keep your code clean and automatically formats it to look professional.

Once installed, you can create a new file, save it as hello.py, and start coding!

Part 2: A Guide to All Necessary Installations – Python via Anaconda

1) Introduction

Another common way to get started with Python, especially in data science, is the **Anaconda** distribution. Instead of installing tools individually, Anaconda is an all-in-one package that contains everything you need right out of the box.

It provides:

- Python itself.
- **Conda**, a package and environment manager that simplifies installing complex libraries.
- Pre-installed IDEs like Jupyter Lab and Jupyter Notebook.
- Hundreds of pre-installed Python libraries tailored for data analysis and machine learning (like Pandas, NumPy, and Scikit-Learn).

The Difference & Our Recommendation: While Anaconda is incredibly convenient because it comes with many pre-installed tools, it is also very large and can be slow. A standalone Python installation combined with Visual Studio Code (see the guide above) is much more lightweight and reflects how professional software developers work. **Therefore, we highly recommend using Visual Studio Code** as your primary IDE, even if you choose to install Anaconda. You can easily connect your Anaconda environment directly to VS Code!

2) Anaconda Installation

1. Download the correct installer for your system from anaconda.com/download. The website will automatically detect whether you need the Windows, macOS, or Linux version.
2. Start the graphical installer by double-clicking the downloaded file. It will guide you through the process. The default settings are usually sufficient.
3. After successful installation, search your applications for a program called **"Anaconda Navigator"** and open it.

3) The Right IDE (Connecting to VS Code)

When you open Anaconda Navigator, you will see a dashboard with various applications like Jupyter Lab, Spyder, and others.

Since we recommend **Visual Studio Code**, you don't need to rely on the basic tools provided by Anaconda. If you have VS Code installed on your computer, it will often appear as a tile right inside the Anaconda Navigator. You can simply click **"Launch"** under the VS Code tile.

By launching it this way, VS Code automatically recognizes your Anaconda installation, giving you the best of both worlds: the massive data science libraries of Anaconda combined with the sleek, professional interface of VS Code.